Multi-armed Bandits

Nonassociative Setting

(does not involve learning to act in more than one situation)

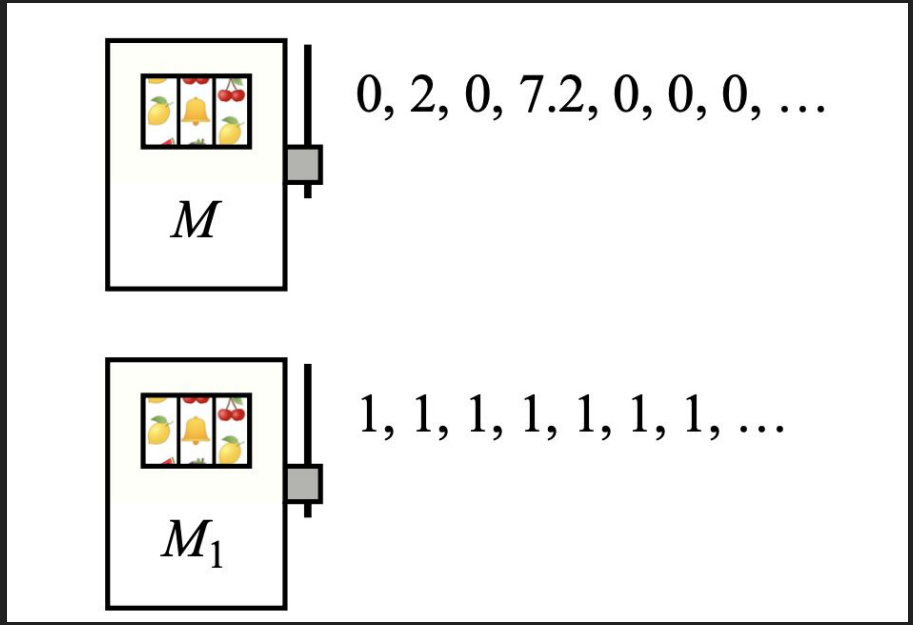
One Armed Bandit

The player receives a reward by pulling the lever.

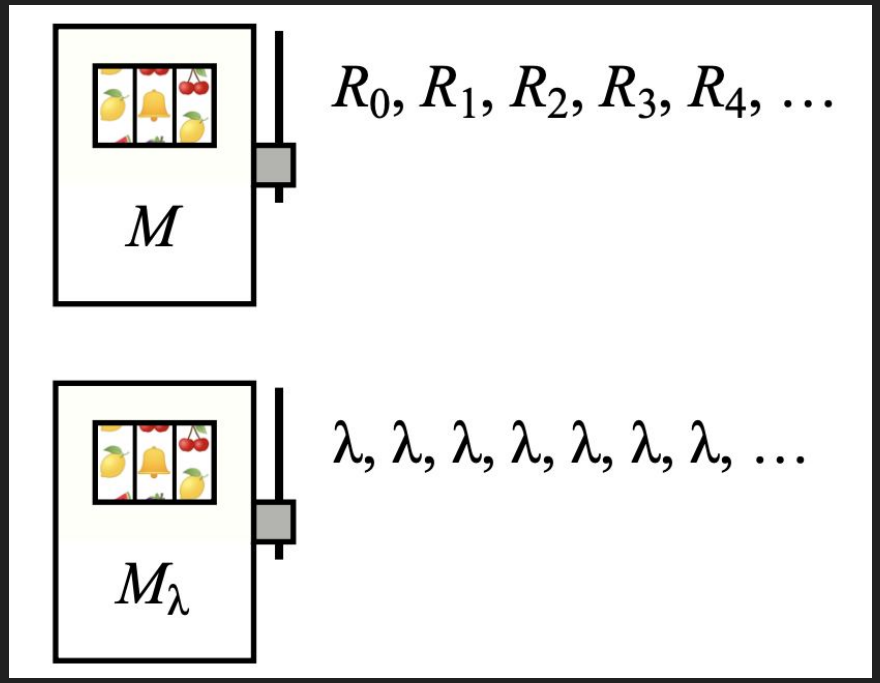
The goal of the player is to maximize the reward over some time period.

The player has no idea how the machine decides rewards. The probability distribution is unknown to the player.

Deterministic One Arm Bandit



Generic One Arm Bandit



Multiple Bandits

K-armed bandits problem offers *k* different options or actions to choose at time *t*.

After each choice you receive a numerical reward from a stationary probability distribution.

The reward depends on the action you selected.

The objective is to maximize the expected total reward over some time period.

Bandit-like Problems

Doctor choosing between experimental treatments for a series of ill patients.

- Actions: choose one of the treatments.
- Reward: survival or well-being of the patient.

- Online Advertisement selection

- A/B Testing of website layouts

Bandit-like Problems

Online Advertisement selection

- Actions: choose which ad (A, B, C, …) to display to a user.
- Unknown reward distribution: click-through rate or conversion probability for each ad.
- Reward: 1 if the user clicks (or converts), 0 otherwise.
- Objective: maximize total clicks over time while learning which ad performs best.

- Each action (ad) has an unknown CTR.
- Each impression is a Bernoulli trial with success probability equal to that CTR.
- The reward is typically 1 for a click and 0 otherwise.

Bandit-like Problems

A/B Testing of website layouts

- Actions: choose which webpage variant (layout, color scheme, headline) to show to each visitor.
- Unknown reward distribution: probability that a visitor completes a desired action (*signup, purchase*).
- Reward: binary or monetary (conversion value).
- Objective: identify and exploit the most effective variant while still exploring alternatives

Value of an Action

Each of the *k* actions has an *expected* or *mean* reward given that the action is selected:

$$q_*(a) = E\,[R_t \mid A_t = a]$$

where

A_t is the action selected at time step t

R_t is the reward corresponding to the action A_t

2.3 The 10-armed Testbed

To roughly assess the relative effectiveness of the greedy and ε-greedy action-value methods, we compared them numerically on a suite of test problems. This was a set of 2000 randomly generated k-armed bandit problems with $k = 10$. For each bandit problem, such as the one shown in Figure 2.1, the action values, $q_*(a)$, $a = 1, \ldots, 10$,

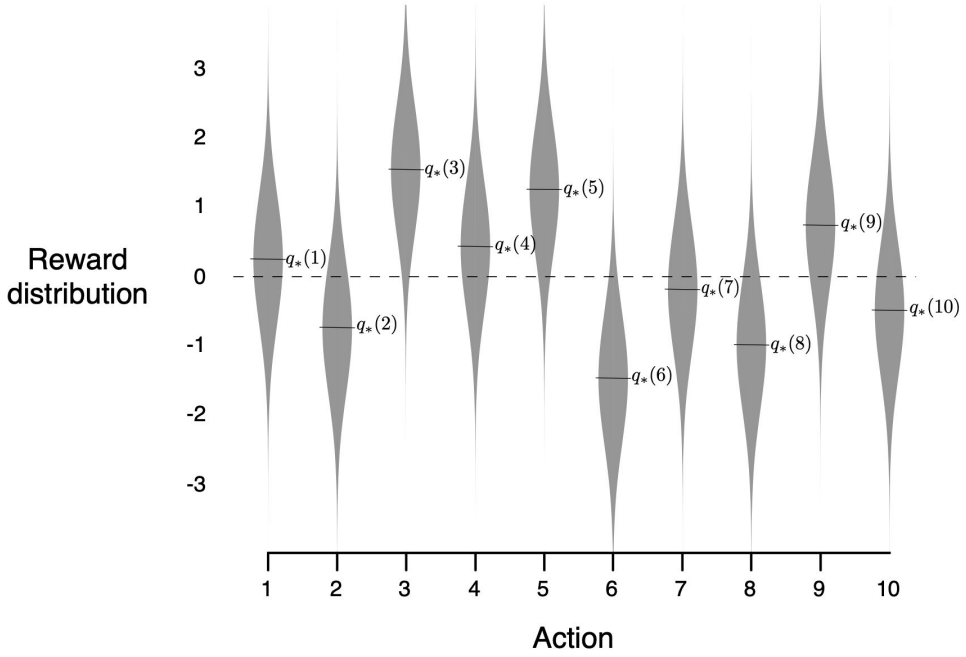


Figure 2.1: An example bandit problem from the 10-armed testbed. The true value $q_*(a)$ of each of the ten actions was selected according to a normal distribution with mean zero and unit variance, and then the actual rewards were selected according to a mean $q_*(a)$ unit variance normal distribution, as suggested by these gray distributions.

- Action A_t applied at times step t
- The reward R_t for action A_t selected from a normal distribution with mean $q_*(A_t)$ and variance 1.
- Choose a learning method
- Measure its performance and behavior as it improves with experience over 1000 time steps

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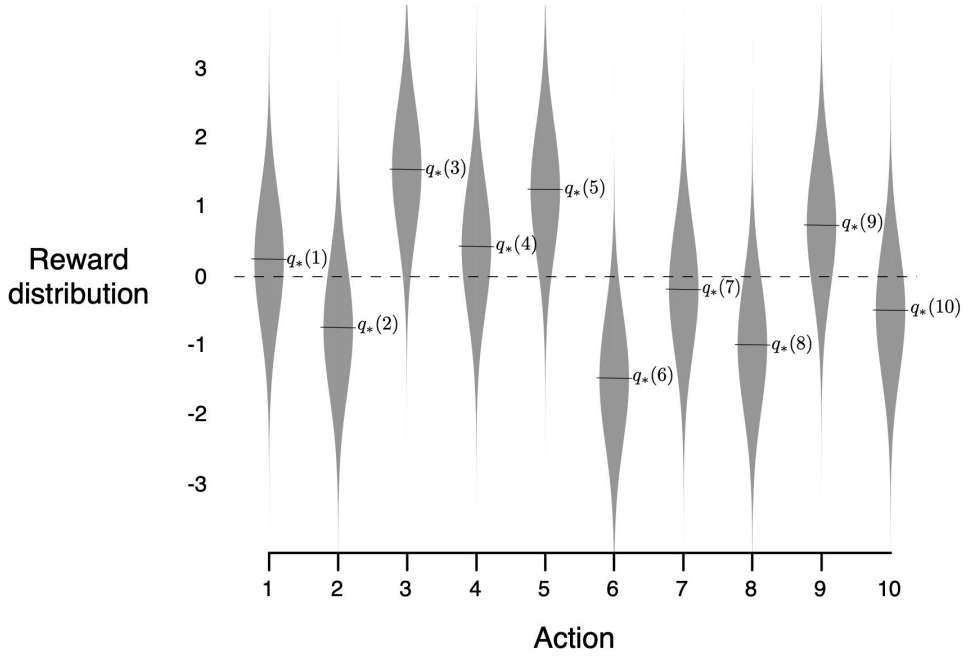

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(1) Is the behavior of one arm causally dependent on some other? Are arms somehow dependent on one another?

(2) Does the reward in time step t_{i+1} depend on the arm pulled in time step t_i?

(3) Are the arms a collection of independent and identically distributed (i.i.d.) variables?

(4) Do the rewards obtained by an arm form an i.i.d?

(5) What do the K arms represent?

Independent and Identically Distributed Variables

The arms are independent.

The arms are *not identically distributed* because they have different *mean* values.

- If they were i.i.d., the question of a “better arm” wouldn’t arise
- The arms provide independent samples from distinct distributions.

The rewards associated with an arm represents a i.i.d.

- For an arm a, at time step t, we sample a new random variable R_t^a

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A_t is the action selected at time step t

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Q: do we know the value of each action? Do we know anything at all?

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Q: do we know the value of each action? Do we know anything at all?

A: if so, we would always select the action with highest value!

Value of an Action and *Estimated* Value of an Action

Each of the *k* actions has an *expected* or *mean* reward given that the action is selected:

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We may have some estimate of the values of $Q_t(a)$

We would like $Q_t(a)$ to be close to $q_*(a)$

Greedy Actions

At any time step there is at least one action whose estimated value is greatest.

These are called *greedy* actions.

When we select a greedy action, we are *exploiting* the knowledge of the *values* of actions.

When we select one of the *nongreedy* actions, we are *exploring*.

Exploration and Exploitation

Conflict between exploration and exploitation.

- Exploration is necessary to *improve* the estimate of nongreedy action's value.
- Exploration is necessary because of uncertainty in estimated values.

- Exploration *may* produce greater total reward in the long run.
- Exploration *may* help discover better actions over many time steps.

Action-Value Methods

Estimating the values of actions.

Using the estimates to make action selection *decisions*.

$$Q_t(a) \doteq \frac{\text{sum of rewards when } a \text{ taken prior to } t}{\text{number of times } a \text{ taken prior to } t} = \frac{\sum_{i=1}^{t-1} R_i \cdot \mathbb{1}_{A_i=a}}{\sum_{i=1}^{t-1} \mathbb{1}_{A_i=a}}, \tag{2.1}$$

where $\mathbb{1}_{predicate}$ denotes the random variable that is 1 if *predicate* is true and 0 if it is not.

Initially, $Q_t(a)$ is set to zero or some default value.

Estimating Action-Value: *sample-average method*

$$Q_t(a) \doteq \frac{\text{sum of rewards when } a \text{ taken prior to } t}{\text{number of times } a \text{ taken prior to } t} = \frac{\sum_{i=1}^{t-1} R_i \cdot \mathbb{1}_{A_i=a}}{\sum_{i=1}^{t-1} \mathbb{1}_{A_i=a}}, \tag{2.1}$$

where $\mathbb{1}_{predicate}$ denotes the random variable that is 1 if *predicate* is true and 0 if it is not.

Initially (*when the denominator is zero*), $Q_t(a)$ is set to 0 or some default value.

As the denominator goes to infinity, $Q_t(a)$ converges to $q_*(a)$

- *the law of large numbers*

Greedy Action Selection Decision

Select one of the actions with the highest estimated value

$$A_t = argmax_a \, Q_t(a)$$

$argmax_a$ denotes the action a for which the expression Q_t is maximized.

ϵ-Greedy Method

Behave greedily most of the time, but every once in a while, with small probability ϵ, select randomly from among all actions with equal probability, independent of the action-value estimates.

Eventually, as the number of steps increase, every action will be sampled an infinite number of times.

This ensures that all the $Q_t(a)$ converges to their respective $q_*(a)$

The advantage of ε-greedy over greedy methods depends on the task. For example, suppose the reward variance had been larger, say 10 instead of 1. With noisier rewards it takes more exploration to find the optimal action, and ε-greedy methods should fare even better relative to the greedy method. On the other hand, if the reward variances were zero, then the greedy method would know the true value of each action after trying it once. In this case the greedy method might actually perform best because it would soon find the optimal action and then never explore. But even in the deterministic case there is a large advantage to exploring if we weaken some of the other assumptions. For example, suppose the bandit task were nonstationary, that is, the true values of the actions changed over time. In this case exploration is needed even in the deterministic case to make sure one of the nongreedy actions has not changed to become better than the greedy one. As we shall see in the next few chapters, nonstationarity is the case most commonly encountered in reinforcement learning. Even if the underlying task is stationary and deterministic, the learner faces a set of banditlike decision tasks each of which changes over time as learning proceeds and the agent's decision-making policy changes. Reinforcement learning requires a balance between exploration and exploitation.

Incrementally Updating the Action Value

Given Q_n is the estimate of the action value after the action has been selected n-1 times

$$Q_n = (R_1 + R_2 + \ldots + R_{n-1}) / (n-1)$$

Given n^{th} reward, the new average of all n rewards can be incrementally computed.

$$\begin{aligned}
Q_{n+1} &= \frac{1}{n}\sum_{i=1}^{n} R_i \\
&= \frac{1}{n}\left(R_n + \sum_{i=1}^{n-1} R_i\right) \\
&= \frac{1}{n}\left(R_n + (n-1)\frac{1}{n-1}\sum_{i=1}^{n-1} R_i\right) \\
&= \frac{1}{n}\Big(R_n + (n-1)Q_n\Big) \\
&= \frac{1}{n}\Big(R_n + nQ_n - Q_n\Big) \\
&= Q_n + \frac{1}{n}\Big[R_n - Q_n\Big],
\end{aligned}$$

A Simple Running Example

n	R	Q
1	$R_1 = 1.4$	$Q_1 = 0$
2	$R_2 = 0.8$	$Q_2 = 1.4/1$
3	$R_3 = 0.3$	$Q_3 = 2.2/2$
4	$R_4 = 0.6$	$Q_4 = 2.5/3$
5	$R_5 = 0.9$	$Q_5 = 3.1/4$
6	$R_6 = -10$	$Q_5 = 4.0/5$

Exercise:

Calculate Q_6 evaluating the incremental update expression.

$$Q_6 = Q_5 + (R_5 - Q_5)/5$$